TALRAG: A Self-Learning Temporal-Adaptive RAG Framework

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Abstract. Retrieval-Augmented Generation(RAG) has become a popular method for reducing hallucinations in Large Language Models(LLMs). However, current RAG frameworks usually rely on static knowledge bases and basic semantic retrieval, which limits their ability to handle domain-specific reasoning and update with new information. This limitation is a major issue in historical question answering systems, where temporal context, causal links, and factual reliability are highly important. To address this, this paper proposes TALRAG, a self learning and temporal adaptive RAG framework built specifically for Vietnamese Historical Question Answering. The system combines adaptive retrieval using multiple scores, self learning guided by trust assessment, and a human-in-the-loop mechanism into a single architecture to keep the knowledge evolving. Unlike standard RAG models, TALRAG continuously updates its knowledge base through a learning pipeline guided by feedback that involves web information gathering, reliability checking, user validation, and automatic vector ingestion. To improve how the model reasons about history, we introduce an retrieval mechanism based on query intent that dynamically scores and combines semantic relevance, time proximity, and causal relationships based on what the user is asking. The whole process runs through a pipeline with multiple stages, including FAISS-based retrieval from multiple vector stores, LLM-based document grading, a web fallback for missing data, and automatic knowledge refinement. Additionally, our learning approach based on user feedback allows new external historical data to become trusted vector memory once validated by users, which improves future search quality and cuts down on hallucination risks. Our experimental results show that TALRAG clearly outperforms traditional static RAG architectures in retrieval robustness, answer relevance, and continuous knowledge adaptation. Overall, this proposed framework offers a scalable and reliable solution for domain-specific AI assistants that require trustworthy reasoning and the ability to learn over time.

Keywords: Retrieval-Augmented Generation(RAG); Self-Learning RAG; Human-in-the-Loop Learning; Temporal Adaptive Retrieval; Historical Question Answering; Knowledge Evolution; Large Language Models.

1 INTRODUCTION

In recent years, the rapid growth of Artificial Intelligence (AI) and digital information has increased the demand for question answering (QA) systems across many domains. Compared with traditional keyword-based search, modern QA systems provide more natural and context-aware access to &information(Riloff&Thelen, 2000). The development of Transformer-based models(Vaswani et al., 2017), including BERT (Devlin et al., 2019) and GPT-4(OpenAI, 2023), has further improved natural language understanding and generation.

Despite these advances, Large Language Models(LLMs) still suffer from hallucination, where generated outputs may contain unsupported or incorrect information(Ji et al., 2023). This issue is especially critical in historical question answering because historical answers require factual accuracy, temporal consistency, and reliable causal interpretation. Therefore, a historical QA system must not only retrieve relevant facts but also reason correctly about dates, dynasties, events, and cause and effect relationships. Retrieval-Augmented Generation(RAG) reduces hallucination by grounding LLM responses in external knowledge sources rather than relying only on model parameters(Lewis et al., 2020; Gao et al., 2024). However, conventional RAG still has limitations in historical QA because it often relies mainly on semantic similarity, lacks explicit temporal and causal reasoning, and usually operates on static knowledge bases that cannot learn from user feedback(Dhingra et al., 2022; Asai et al., 2023; Yan et al., 2024). To address these limitations, this paper introduces TALRAG, a RAG framework with self learning and temporal adaptation for Vietnamese historical question answering. Unlike static RAG models, TALRAG learns from validated user feedback and updates its vector memory through a self learning pipeline guided by trust assessment.

The rest of this paper is organized as follows. Section 2 reviews related work on RAG, adaptive retrieval, and self learning QA systems. Section 3 presents the proposed TALRAG framework. Section 4 describes the experimental setup and results. Section 5 concludes the paper and discusses future work.

2 RELATED WORK

The development of question answering(QA) systems has evolved from rule-based methods to neural and retrieval-augmented approaches. Early QA systems relied mainly on handcrafted rules and keyword matching, which provided controllability but limited contextual reasoning &capability(Riloff&Thelen, 2000; Lehnert, 1977). The emergence of Transformer-based models significantly improved natural language understanding and generation(Vaswani et al., 2017). Pre-trained models such as BERT and GPT-4 further advanced contextual representation, reasoning, and conversational interaction(Devlin et al., 2019; OpenAI, 2023). However, LLMs still suffer from hallucination, where generated answers may contain unsupported or factually incorrect information (Ji et al., 2023). This limitation is especially critical in historical QA, where factual accuracy, temporal consistency, and causal interpretation are essential.

Retrieval-Augmented Generation(RAG) addresses hallucination by grounding LLM outputs in external documents(Lewis et al., 2020). Dense retrieval methods such as DPR and Sentence-BERT improve semantic search, while FAISS enables efficient vector retrieval over large-scale document collections(Karpukhin et al., 2020; &Reimers&Gurevych, 2019; Johnson et al., 2019). Recent surveys show that RAG has become an important approach for knowledge-intensive NLP tasks(Gao et al., 2024). Nevertheless, conventional RAG systems still face limitations in historical question answering. They often rely mainly on semantic similarity, which may retrieve documents that are linguistically related but temporally or causally inconsistent. Prior work on time-aware language models highlights the importance of temporal reasoning for knowledge-intensive tasks involving timelines and evolving facts (Dhingra et al., 2022).

Several studies have attempted to improve retrieval quality and answer reliability. BERT-based re-ranking improves passage ranking by refining the order of retrieved documents(Nogueira&Cho, 2019), while BEIR provides a heterogeneous benchmark for evaluating retrieval models across domains (Thakur et al., 2021). More recent frameworks such as Self-RAG and Corrective RAG introduce self-reflection or correction mechanisms to improve generation quality(Asai et al., 2023; Yan et al., 2024). GraphRAG further explores graph-based retrieval to capture global relationships among entities(Edge et al., 2024). However, these approaches still do not fully address the need for a transparent mechanism that continuously updates domain knowledge from validated user interaction history.

In the Vietnamese NLP domain, Nguyen and Nguyen(2020) introduced PhoNLP, which supports Vietnamese language processing and is relevant to historical QA because Vietnamese historical questions often contain named entities, dynastic names, temporal expressions, and culturally specific terminology. For the historical QA baseline, this study uses ItihashQA. ItihashQA is a retrieval-augmented conversational question answering system developed for the Bangladeshi historical context. Its released implementation follows a static RAG architecture, where retrieved historical contexts are passed to a language model for response generation. Since ItihashQA is specifically designed for historical question answering, this study adopts it as the baseline system and adapts its static RAG pipeline to the Vietnamese historical corpus for controlled comparison. Although the repository description also refers to the system as EkattorQA, this paper consistently uses the name ItihashQA according to the repository and work title.

Commercial document-grounded QA tools such as Google NotebookLM and Gemini Gems also demonstrate the practical value of retrieval-based assistants for document understanding and personalized knowledge access. These systems allow users to organize sources or create customized assistants for specific tasks. However, their retrieval mechanisms, evaluation procedures, and memory update strategies are not fully transparent for research reproduction. More importantly, they do not provide an explicit self learning pipeline guided by trust assessment, in which chat history validated by users is converted into new vector memory. Therefore, although these tools are useful for document synthesis and personalized assistance, they are not directly equivalent to TALRAG's knowledge evolution mechanism guided by feedback.

From the reviewed literature, three limitations remain clear. First, most RAG systems rely heavily on semantic similarity and do not explicitly optimize retrieval for temporal and causal reasoning, which are essential in historical QA. Second, many adaptive RAG frameworks improve retrieval or generation quality but still operate on static knowledge bases after deployment. Third, existing document-grounded chatbot systems do not provide a transparent research-oriented mechanism for converting user feedback and chat history into validated vector memory.

TALRAG addresses these gaps by combining retrieval based on query intent, temporal and causal scoring, document grading using an LLM, web fallback for missing information, and self learning guided by feedback. The key novelty of TALRAG is its ability to learn from user chat history. When the system encounters a low-confidence or unanswered query, the interaction becomes a learning candidate. After external knowledge acquisition, reliability verification, and positive user feedback, the validated knowledge is refined and ingested into the dynamic FAISS vector store. In this way, TALRAG does not simply answer questions from a fixed corpus; it continuously evolves its historical knowledge base through knowledge validation guided by feedback.

3 PROPOSED TALRAG FRAMEWORK

The main contribution of TALRAG is a self learning pipeline guided by trust assessment that converts low-confidence or unanswered queries into opportunities for knowledge expansion. Unlike standard RAG systems that rely on fixed vector databases, TALRAG uses validated user feedback and chat history to update its dynamic vector memory over time.

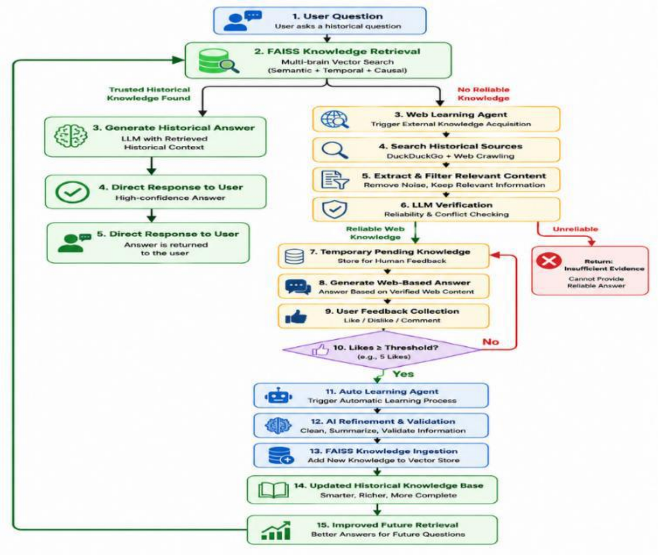


Fig. … AG.

To forma … main inference … retrieval, docume … learning pipeline b … al failures are conve … , reliability

verification, …

Algorithm 1 …

Input: Use… …es V, language m… … C

1: Authent…

2: Check u…

3: intent ← …

4: temporal_info ←ExtractTemporalEntities(q)

5: q_norm ←NormalizeQuestion(q)

6: candidate_docs ←MultiBrainFAISSRetrieve(q_norm, V)

7: for each document d in candidate_docs do

8: S_sem ←SemanticScore(q_norm, d)

9: S_temp ←TemporalScore(temporal_info, d)

10: S_caus ←CausalScore(q_norm, d) 11: S_total ←α(intent) × S_sem + β(intent) × S_temp + γ(intent) × S_caus

12: end for

13: ranked_docs ←SortByScore(candidate_docs, S_total)

```
filtered_docs ←DocumentGrader(q, ranked_docs)
if filtered_docs is not empty then
    C ←filtered_docs
    A ←GenerateAnswer(LLM, q, C, H)
else
    A, C ←TrustAwareSelfLearning(q)
end if
Save conversation log
Update token balance
return A, C
```

Algorithm 2. Trust-aware self-learning pipeline
Input: User question q, web search engine W, language model LLM, pending knowledge repository P, dynamic vector store V_dynamic, feedback threshold θ**Output**: Provisional answer A_web, verified web contexts C_web

```
web_results ←SearchWeb(W, q)
raw_chunks ←CrawlAndExtract(web_results)
relevant_chunks ←FilterRelevantContent(q, raw_chunks)
C_web, confidence ←VerifyReliability(LLM, q, relevant_chunks)
if confidence is low then
A_web ←"Insufficient evidence"
return A_web, ∅
end if
A_web ←GenerateWebBasedAnswer(LLM, q, C_web)
SaveToPendingKnowledge(P, q, A_web, C_web)
Collect user feedback for q
if PositiveFeedback(q) ≥θ then
    refined_knowledge ←RefineKnowledge(LLM, q, A_web, C_web)
    chunks ←ChunkText(refined_knowledge)
    embeddings ←GenerateEmbeddings(chunks)
    IngestIntoFAISS(V_dynamic, chunks, embeddings)
    MarkKnowledgeAsApproved(P, q)
end if
return A_web, C_web
```

The TALRAG workflow begins when a user submits a historical question. The system first performs authentication and token balance checking to ensure secure and controlled access. The query is then classified into intent types such as factual, temporal, causal, or comparative questions. This classification guides the retrieval process because different historical questions require different evidence priorities.

To support historical reasoning, TALRAG applies a lightweight temporal extraction mechanism to identify years, dynasties, historical periods, and other time-related expressions. These temporal signals are used to guide document retrieval and ranking, helping the system avoid contexts that are semantically similar but historically misaligned.

The retrieval component is based on multi-source FAISS retrieval. It retrieves candidate documents from trusted historical sources, approved knowledge, and dynamically learned vector memory. Instead of ranking documents only by semantic similarity, TALRAG computes a combined retrieval score:

$$S_{total} = (a \times S_{semantic}) + (b \times S_{temporal}) + (\gamma \times S_{causal})$$

where $S_{semantic}$ measures semantic similarity between the question and the document, S_temporal measures chronological consistency, and S_{causal} measures causal alignment with the query. The weights α, β, and γ are adjusted according to the classified query intent. For example, temporal questions assign higher weight to $S_{temporal}$, while causal questions prioritize S_{causal}.

After retrieval, the ranked documents are processed by an LLM-based document grading module. This module filters noisy, weakly related, or contradictory contexts before answer generation. If sufficient reliable context remains, the LLM generates a grounded answer using the filtered evidence.

If the trusted retrieval pipeline cannot find sufficient reliable knowledge, TALRAG activates the trust-aware self-learning pipeline. In this stage, the system searches external sources and extracts candidate historical evidence. The extracted content is filtered for relevance and checked for reliability before being used to generate a provisional answer. Importantly, knowledge acquired from the web is not directly inserted into the trusted knowledge base. Instead, it is stored in a pending knowledge repository until it receives sufficient positive user feedback.

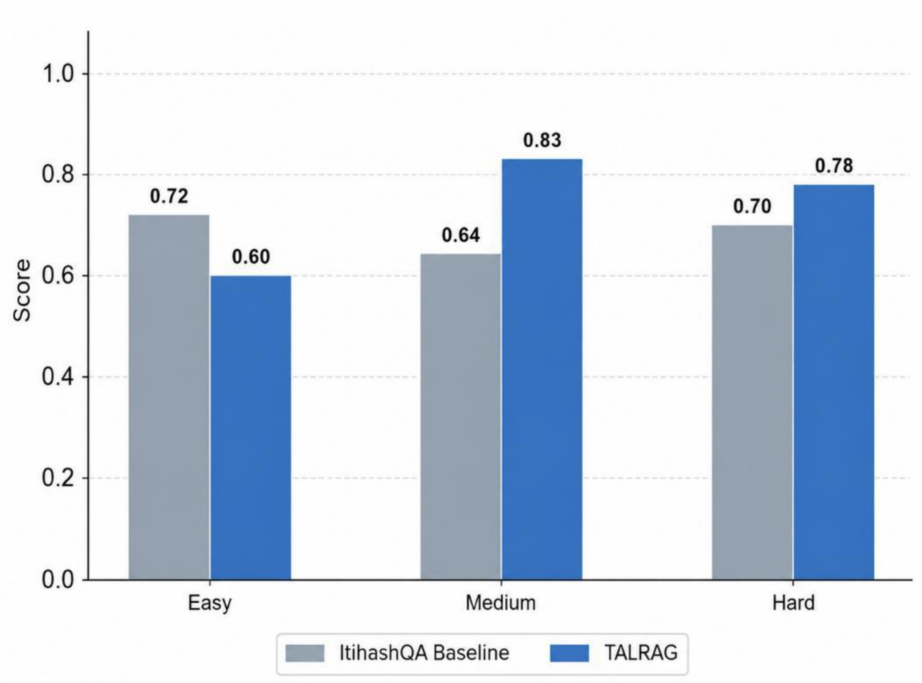
The central novelty of this framework is the use of user interaction history as a learning signal. When a provisional answer receives enough positive feedback, the automatic learning module refines the verified content, splits it into chunks, generates embeddings, and ingests it into the dynamic FAISS vector store. As a result, future similar questions can retrieve this validated knowledge directly from vector memory.

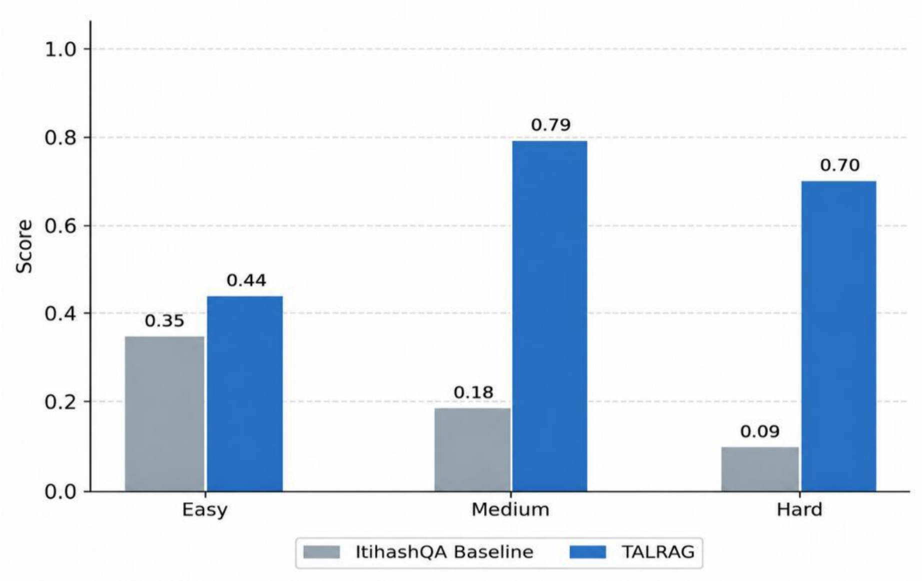
This design allows TALRAG to continuously improve its historical coverage while reducing the risk of contaminating the trusted knowledge base with unreliable information. Finally, the system records interaction logs, retrieval traces, feedback signals, and token usage for monitoring, resource management, and future analysis.

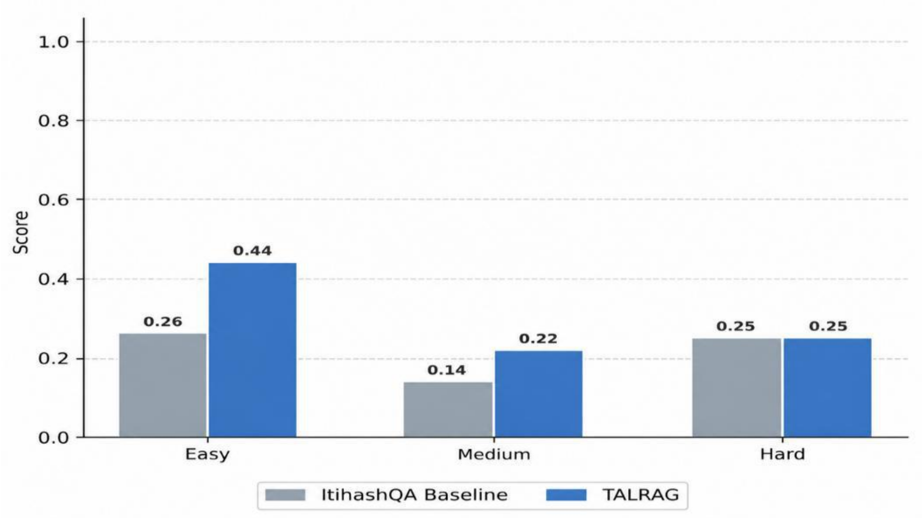
4 EXPERIMENTS

To evaluate the effectiveness of TALRAG, we conducted a RAGAS-based comparative experiment against the ItihashQA baseline. ItihashQA is a retrieval-augmented historical question answering system originally developed for the Bangladeshi historical context. In this study, its static RAG pipeline is adapted to the Vietnamese historical corpus for controlled comparison. The baseline retrieves the top-k semantically similar chunks from the vector database and directly passes them to the language model for answer generation.

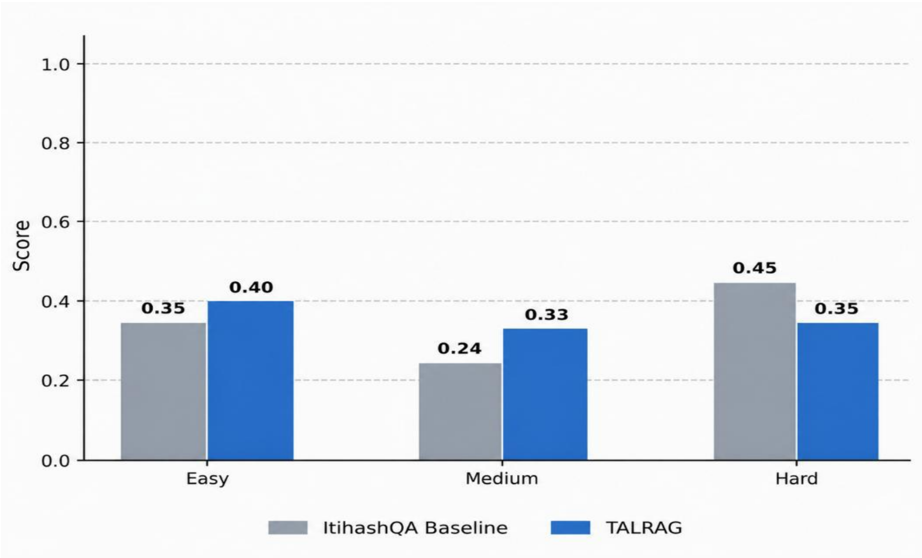
In contrast, TALRAG extends the standard RAG workflow with intent classification, temporal and causal scoring, document grading using an LLM, web


1.0
0.8
0.6
0.4
0.2
0.0
Score
0.72
0.60
0.64
0.83
0.70
0.78
Easy
Medium
Hard
ItihashQA Baseline
TALRAG


1.0
0.8
0.6
0.4
0.2
0.0
Score
0.79
0.70
0.44
0.35
0.18
0.09
Easy
Medium
Hard
ItihashQA Baseline
TALRAG


1.0
0.8
0.6
0.4
0.2
0.0
Score
0.44
0.26
0.22
0.14
0.25
0.25
Easy
Medium
Hard
ItihashQA Baseline
TALRAG

In terms of Context Precision, TALRAG improves over the baseline across all difficulty levels. On Easy questions, the score increases from 0.26 to 0.44. On Medium questions, it increases from 0.14 to 0.22. On Hard questions, both systems achieve approximately 0.25, with TALRAG showing a slight improvement. These results show that temporal-causal scoring and document grading help retrieve cleaner and more relevant contexts.



LLM-based document grading, and the self learning pipeline guided by trust assesment. At the same time, the results show that TALRAG does not outperform the baseline in every individual case. In particular, the lower Easy Faithfulness score and lower Hard Context Recall score suggest that stricter filtering can reduce useful evidence coverage. Future work should investigate adaptive thresholds, larger top-k retrieval for difficult questions, and better precision-recall balancing.

The RAGAS evaluation shows that TALRAG provides a more relevant, context-aware, and RAG architecture that can improve over time for Vietnamese historical QA. By combining temporal and causal retrieval, document grading, and trust-aware self-learning, TALRAG turns static process of retrieval and generation into a continuously evolving historical knowledge system.

5 CONCLUSION

This research introduced TALRAG, a RAG framework with self learning and temporal adaptation for Vietnamese historical question answering. Unlike static RAG systems that mainly rely on semantic similarity and fixed vector databases, TALRAG combines adaptive retrieval, temporal and causal reasoning, document grading, and self learning guided by trust assessment in a unified workflow.

The main contribution of TALRAG is its self learning pipeline guided by trust assessment. When reliable information cannot be retrieved from the trusted vector store, the system acquires supplementary historical evidence from external sources, verifies its reliability, stores it as pending knowledge, and uses user feedback to decide whether it should be ingested into FAISS as reusable vector memory. This mechanism allows TALRAG to learn from user interaction history while reducing the risk of contaminating the knowledge base with unreliable information.

TALRAG demonstrates that RAG can be extended from a static process of retrieval and generation into a continuously evolving historical knowledge system. Future work will focus on improving recall for multi-hop questions, integrating knowledge graphs to model relationships among historical entities, and exploring multimodal retrieval for historical maps, scanned documents, ancient scripts, and artifacts.

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